Bhavin K. Moriya



Data Scientist | Machine Learning | Privacy-Preserving Analytics
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Profile

Data scientist with a **Ph.D. in Mathematics** and 2+ years building production data-science tools in German applied research (BMBF project, Hochschule Esslingen). Rare combination of **statistical modelling, machine learning, and homomorphic-encryption engineering** (OpenFHE/C++, TFHE-rs/Rust). 17 peer-reviewed publications. Builds documented, reproducible tools and explains technical results clearly to specialist and non-specialist audiences. Seeking data science / quantitative modelling roles in industry.

Core Skills

Programming: Python (NumPy, pandas, polars, scikit-learn, PyTorch), SQL, C++, Rust; also R, Julia, Go
ML & statistics Neural networks, RNNs, bagging and boosting, regression and classification, MCMC, causal inference, numerical linear algebra, finite element methods
Privacy tech: Homomorphic encryption (CKKS, TFHE); OpenFHE (C++), TFHE-rs (Rust); encrypted comparison, polynomial evaluation, square root
Quantitative finance: Black–Scholes–Merton-Merton pricing implemented from scratch; Monte Carlo simulation of financial products; numerical solution of SDEs; portfolio optimisation (CAPM, Markowitz); risk modelling; credit-risk modelling (PD / LGD / CCF)
Data & tooling: geopandas, SQLite, PostGreSQL, SUMO, Docker, Git, Gradio, Streamlit, uv, LaTeX
Languages: English (fluent), German (A2, actively working towards B1), Portuguese (fluent), Hindi & Gujarati (native)

Professional Experience

Data Scientist / Research Assistant — **Hochschule Esslingen, Germany** — **Nov 2023 – Dec 2025**

- **AnoMoB (BMBF research project)**: Engineered first-of-its-kind encrypted comparison operations and non-linear function evaluation (square roots, polynomials) using **OpenFHE (C++)** and **TFHE-rs (Rust)** for privacy-preserving mobility analytics.
- **Benchmarking study:** Co-authored a performance comparison of runtime and memory consumption of the **CKKS and TFHE** schemes; published on the Cryptology ePrint Archive (2025).
- **Data engineering:** Built production-quality tools for statistical analysis of large mobility datasets of around 2 million rows using **Python, polars, and geopandas**.
- **Mobility simulation:** Simulated traffic with **SUMO** and compared travel modalities by CO_2 emissions and route duration using OpenRouteService/OpenTripPlanner (Dockerized architecture).
- **Decision-support tool:** Developed a weighted-scoring application (Streamlit) that lets users trade off travel time against CO_2 across transport modes.

Assistant Professor — **Universidade Federal de Viçosa, Brazil** — **Nov 2015 – Nov 2023**

- Taught 29 undergraduate and graduate courses (calculus, linear algebra, algebra, finite mathematics, rings and modules), including to engineering students; lectured in English and Portuguese.
- Supervised 2 master's dissertations and 1 undergraduate research project.
- Led a FAPEMIG-funded international research project; published 16+ peer-reviewed articles.
- Contributed to a FAPEMIG-funded international research project on zero-sum problems; 16+ peer-reviewed publications with collaborators in Brazil, India, China, UK, USA, Germany and France.

- Organised two Summer Schools for students at the university (2017 and 2021–22).

Postdoctoral Fellowships: IMSc Chennai (2011–2013) | HRI Allahabad (2013–2014) | CMI (2014) | University of Brasília (2014–2015)

Education

- **Ph.D. in Mathematics**, Harish Chandra Research Institute 2011
  Thesis: Some Zero Sum Problems in Combinatorial Number Theory
- **M.Sc. & B.Sc. in Mathematics**, The M.S. University of Baroda 2003 – 2005

Selected Projects

- **Credit-Risk Model (PD · LGD · CCF)** — end-to-end probability-of-default, loss-given-default, and credit-conversion-factor modelling pipeline.
- **FEM Solver** — finite-element-method solver written from scratch in Python.
- **AudioMNIST** — spoken-digit recognition system (neural networks, DSP).
- **Audio DSP Web App** — interactive tool exploring the mathematics of audio digital signal processing. Demo
- **Vascular Surrogate Models** — machine-learning surrogates accelerating intracranial-aneurysm CFD.

Selected Publications

- (with C. Krüger, D. Schoop) A Performance Comparison of the Homomorphic Encryption Schemes CKKS and TFHE. *Cryptology ePrint Archive* 2025/1460.
- Signed zero sum problems for metacyclic groups. *Periodica Mathematica Hungarica* (2024).
- (with Martínez, Lemos, Ribas) The main zero-sum constants over $D_{2n} \times C_2$. *SIAM J. Discrete Math.* 37(3), 1496–1508 (2023).
- (with Lemos, Moura, Silva) On the number of fully weighted zero-sum subsequences. *Int. J. Number Theory* 15(5) (2019).
- (with C. J. Smyth) Index-dependent divisors of coefficients of modular forms. *Int. J. Number Theory* 9(7) (2013).
- (with Chintamani, Gao, Paul, Thangadurai) New upper bounds for the Davenport and for the Erdős–Ginzburg–Ziv constants. *Arch. Math. (Basel)* 98(2) (2012).

Full list of 16+ publications: Google Scholar

Talks, Conferences and Awards

- **Conferences:** Privacy Enhancing Technologies Symposium (PETS), Bristol 2024; PET-CON, TU Berlin 2024; AnoSiDat, Lübeck 2024; Anonymization of Integrated and Georeferenced Data, Berlin 2024.
- **Invited talks:** Séminaire CAESAR, Sorbonne (Paris 6) 2012; University of Brasília 2015; Universidade Federal de Viçosa 2016 and 2017; colloquium at IMSc Chennai 2011.
- **Awards:** NBHM grant to attend the International Congress of Mathematicians (Seoul, 2014); NBHM scholarship for Ph.D. studies; CIMPA-ICTP travel grant (2013).

References

Available on request, including **Prof. Dr. Dominik Schoop (Hochschule Esslingen)**, **Prof. Jean-Paul Allouche (CNRS, Sorbonne Université)** and **Dr. Wolfgang Schmid (Département de Mathématiques et H.S., Université Paris 8)**